Math 540 -Linear Algebra II
Fall 2026
HW1 - Vector spaces, Short exact sequences, Dimension
Submit via Canvas by Sunday 9/20/26 23:59pm
(Quiz on it on Monday 9/21/26 in class)

Remarks:

A) Definition is just a definition, there is no need to justify or explain it.

B) Answers to questions with proofs should be written, as much as you can, in the following format:

i) Statement.

ii) Main points that will appear in your proof.

iii) The actual proof.

C) Answers to questions with computations should be written, as much as possible, in the following format:

i) Statement and Result.

ii) Main points that will appear in your computation.

iii) The actual computation.

1. *Vector spaces.* Suppose $\mathbb{F}$ is a field.

   (a) Define when we say that a vector space V over $\mathbb{F}$, is finite-dimensional.

   (b) Consider the vector space
   $$V = \mathbb{F}[x],$$
   of all polynomials with coefficients in $\mathbb{F}$.
   Show that V is not finite-dimensional.

   (c) Suppose X is a finite set. Consider the vector space V, of all functions from X to $\mathbb{F}$,
   $$V = \mathbb{F}(X) := \{\text{all } f : X \to \mathbb{F};\ \text{s.t. } f \text{ is a function}\},$$
   with the standard addition and multiplication by scalars from $\mathbb{F}$.
   Show that V is finite-dimensional.

2. *Short exact sequences.* Suppose U, V, W are three vector spaces over $\mathbb{F}$. Consider the following sequence of spaces and linear transformations between them:
   $$0 \to U \xrightarrow{\iota} V \xrightarrow{\epsilon} W \to 0, \tag{1}$$
   where $0 \to U$ is the obvious map from the zero space into U, and $W \to 0$ is the obvious map from W onto the zero space.

(a) Define when we say that the sequence (1) is short exact sequence (s.e.s.).

(b) Given two subspaces $U, W < V$, such that $V = U \oplus W$, write down a natural s.e.s. of the form (1).

(c) Let Y be a set and $X \subset Y$ be a subset. Show that there is a natural s.e.s. associated with the spaces of functions $U = \mathbb{F}(X)$, $V = \mathbb{F}(Y)$, and $W = \mathbb{F}(Y \setminus X)$, where $Y \setminus X$ denotes set-minus, i.e., the set of elements which are in Y and not in X

3. *Dimension.* Denote by $Vect_{\mathbb{F}}^{fd}$ the collection of finite-dimensional vector spaces over $\mathbb{F}$, with linear transformations between them.

(a) State the fact about uniqueness and existence of unique dimension function

$$\dim : Vect_{\mathbb{F}}^{fd} \to \mathbb{N},$$

that satisfies certain desired properties.

Def. For V finite dimensional, the integer $\dim(V)$ is called the dimension of V.

(b) Show that $\dim(M_n(\mathbb{F})) = n^2$.

(c) Suppose $1 + 1 \neq 0$ in $\mathbb{F}$. Consider the spaces $U = A_n(\mathbb{F})$, $V = M_n(\mathbb{F})$, $W = S_n(\mathbb{F})$, of anti-symmetric matrices (satisfy $A^t = -A$), all matrices, and symmetric matrices (satisfy $A^t = A$), respectively.

1. Show that, they form in a natural why a s.e.s.
2. Deduce that: $\dim(A_n(\mathbb{F})) = \frac{n \cdot (n-1)}{2}$, and $\dim(S_n(\mathbb{F})) = \frac{n \cdot (n+1)}{2}$.

Important remarks:

- Every Tue-Thu 8:30-9pm we go (only if students show up to the meeting) over HW in canvas's zoom (and record it).
- You are very much encouraged to consult the lecturer and grader on HW during office hours/appointments/discussions/lectures.
- You are very much encouraged to work with other students on the HW.
- You should submit your HW **ALONE in YOUR OWN ORIGINAL DOCUMENT!!!**
- Remember: "ChatGpt" will not be there for you during the quiz on the HW, so make sure you can really solve the HW problems.

Good Luck!